Group 5
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Optogenetic Intervention as a Potential Treatment for Borderline Personality Disorder

I. Introduction

Borderline Personality Disorder (BPD) remains one of the most challenging psychiatric conditions to treat (Biskin & Paris, 2012). Characterized by rapidly fluctuating moods, impulsivity, and difficulty maintaining interpersonal relationships, the lifetime prevalence of BPD is between 0.7% to 2.7% in the general adult population (Leichsenring et al., 2024). Current mainstream treatments for BPD such as dialectical behaviour therapy (DBT) require long-term commitment, while medications can help with co-occurring conditions like depression or anxiety but fail to directly treat the core mechanisms of BPD. Even with these existing treatments, diagnostic remission occurs in 85% to 93% of patients over 10 years (Soloff, 2021), highlighting the opportunity for more effective and lasting BPD treatments.

Recent breakthroughs in neuroscience have shown that optogenetics—a biological technique using a combination of light and genetic engineering to control the activity of cells—may be used to activate, neutralize, and modify specific memories (Liu et al., 2012). Through optogenetics studies with mice, targeting hippocampal neurons associated with fear memories and reactivating these circuits in a neutral context has been shown to rewire the subject's emotional response to those initial fear-related memories. If this could be translated to humans, by targeting the neural networks underlying emotional memory, optogenetics could potentially be more durable than traditional therapies that rely on behaviour alone.

Our project proposes a hypothetical intervention utilizing optogenetics to treat individuals with treatment-resistant BPD, by comparing outcomes between a standard control group and an optogenetic intervention group. This research draws on psychology, by investigating how BPD manifests in the human brain and exploring a potential new treatment for humans using optogenetics, as well as philosophy, by examining the implications of tampering with the nature of selfhood and the extent to which memories make up human identity.

II. Background Research

While BPD has been widely studied in psychology, current treatments rely primarily on psychotherapy, or "talk therapy", which concerns the treatment of mental conditions through verbal communication and interaction (Bateman & Fonagy, 2019). Approaches such as DBT—the current gold standard for BPD intervention—focus on behavioural change and long-term coping development. However, such interventions lack the ability to directly modify the neural circuits responsible for emotional instability and maladaptive memory processing present in BPD. And as a result, even after remission, full recovery remains difficult to achieve. Although many individuals experience long symptom-free periods, only about 50% attain full recovery over a 10-year period, and residual symptoms often persist (Zanarini et al., 2010).

Additionally, research about optogenetics on human participants for psychiatric purposes is sparse, and has yet to be explored extensively due to its potential technical limitations and ethical concerns. Related neuromodulation methods such as deep brain stimulation and transcranial magnetic stimulation (Lozano et al., 2019; George & Post, 2011)—which use electrodes or magnetic pulses to stimulate brain circuits—are already used clinically for conditions like Parkinson's disease, OCD, and depression. While these treatments provide proof of concept that invasive neuromodulation is possible in humans, they differ from optogenetics in that they are much less precise and cannot target memory encoding cells. Therefore, we will build on these existing neuromodulation studies, using BPD as a compelling candidate for examining whether a refined approach like optogenetics could more effectively target the emotional memory processes central to this disorder.

III. Methodology

a. Sampling and Procedure

A voluntary sample of participants must meet the following inclusion criteria: a) clinically diagnosed with BPD based on the DSM-5-TR, which standardizes psychiatric diagnoses, b) above the age of 18, and c) absence of significant immune or chronic illnesses and conditions that could pose risks during neuromodulation research. This criteria ensures the safety of patients during research.

We will recruit around 24 clinically diagnosed BPD participants. As this is an early-phase human trial study, this sample size limits participant involvement to just what is necessary to acquire sufficient statistical power needed to detect the effectiveness of the treatment, while balancing safety given our technique's potentially invasive nature. Resources required for high-quality and tested optogenetics equipment, as well as continuous monitoring of the patients to ensure procedural safety will also be costly, which will constrain our sample size.

Prior to experimentation, participants are to complete clinical interviews and BPD-specific self-report scales (eg. BPDSI, BSL-23) in order to determine initial BPD symptoms and severity. During these interviews, the specific emotionally significant memory of interest will be determined for each individual, which will later serve as the target for the experimental condition.

Participants will be randomly allocated to either the control condition or the experimental condition, where they will not be told the nature of their condition. The control group will receive a placebo treatment with no changes to their memory. Conversely, the experimental group will undergo optogenetic treatment. Both conditions will receive treatment over the course of 4 months, allowing for meaningful long-term monitoring.

After the treatment stage, participants are to complete another round of clinical interviews and self-reported scales. During this stage, researchers will be unaware of the participant's condition during prior stages of the research. Hence, this is a double-blind experiment. Employing this structure will mitigate potential researcher biases and participant demand characteristics. Comparing the results from this stage to the initial screening will provide findings as to whether optogenetics produces measurable improvements in treating BPD.

To adhere to ethical standards, participants will be provided a debriefing form that includes information about the research and the option to withdraw their data following the experiment. Additionally, all participants will be monitored for 6 months after the experiment for any significant side effects.

b. Philosophical Discussion

Optogenetic treatment actively targets an individual's memories, raising questions of the role of past experiences in establishing a cohesive self identity. The limitations imposed on optogenetics is dependent on the philosophical theory one concerns themselves with in regard to human identity. This paper will delve into a number of existing theoretical stances, specifically those pertaining to the mind and body problem.

Following the functionalist school of thought, individual memories play a larger functional role in the mental economy to develop a person's character traits and identity (Heil, 2004). Since functionalists do not define single memories based on their distinct functionality, removal of a specific memory of interest would have the same impact as removal of other memories. Therefore, tampering with participant's past experiences would have cascading implications for the participants' self. On the other end of the spectrum, type identity theorists may argue that a single instance of one specific memory corresponds to one aspect of an individual's identity (Chen, 2023). In that respect, performing optogenetics specified at the memory of interest will have limited effects outwards to the participants' self identity as a whole.

Similar to how the functionalist story raises a large concern for optogenetic treatment, the behaviourist reinforces these concerns. Behaviourists may argue that memories act as inputs that elicit certain outputs, expressed through behaviour (Leeder, 2022). In that vein, tampering with one instance of memory could mean uprooting a pillar of behaviour.

IV. Discussion

Our proposed methodology integrates empirical psychological evaluation with philosophical questioning in order to evaluate both the clinical value and broader implications of optogenetic memory intervention for BPD. We chose this interdisciplinary approach because BPD involves both neural and psychological aspects—in particular, emotional dysregulation and maladaptive memory processing—which, based on current evidence, cannot be fully addressed by behavioural therapies alone. We also chose to carefully consider the philosophical and ethical implications because memory-targeted interventions to an optogenetic degree have never been applied to humans, and considering this perspective helps us to evaluate our study's potential impact on individuals and society more broadly.

Based on previous work demonstrating successful modulation of emotional memory networks in animal models, we anticipate that participants receiving optogenetic treatment will show a greater reduction in BPD symptom severity relative to the control group. A statistically significant decrease in BPDSI and BSL-23 scores—relative to the control group, and consistent across multiple measures—following intervention would provide preliminary evidence that altering the neural circuits underlying traumatic memory can alleviate core BPD symptoms, including impulsivity and emotional instability (Arntz et al., 2003; Bohus et al., 2009).

Conversely, if results show limited or inconsistent improvement, this would suggest that BPD's highly variable nature—where symptoms of the disorder can differ significantly between individuals—and its complex developmental origins cannot be sufficiently addressed through interventions targeting a single memory network. This would align with previous findings indicating that BPD arises from various psychological and neural factors (Leichsenring et al., 2024).

The feasibility of optogenetics in human psychiatric treatment remains uncertain. While optogenetics provides precise cellular control in mice, isolating a single traumatic memory in humans is far more complicated (Kohara & Okada, 2023), and the invasive nature of the procedure raises practical risks for human application (Goshen, 2014). However, given the demonstrated success of related neuromodulation techniques in humans, and the controlled procedures included in our proposal, we are optimistic about the potential applicability of this approach.

Regardless of outcome, our study would advance psychological research by empirically testing whether memory-centered neuromodulation such as optogenetics can bridge the gap between remission and full recovery. At the same time, engaging philosophical frameworks around identity and autonomy highlights that the ethical stakes of optogenetic intervention expand beyond clinical benefit.

Finally, societal risks must also be acknowledged. Potential consequences include misuse of memory modification technologies, misdiagnosis leading to inappropriate intervention, and unpredictable long-term psychological effects. Future research must therefore address not only clinical feasibility but also regulatory structures, careful supervision, and equitable access. Nevertheless, designing a controlled experiment like the one proposed here provides a conceptual foundation that can guide future research and the responsible development of these technologies as they advance.

V. Conclusion

Our project proposes a novel interdisciplinary approach to treating BPD by applying optogenetic memory-modification techniques to target the neural mechanisms underlying emotional dysregulation. Drawing on neuroscientific research on memory modulation and philosophical inquiry into identity and autonomy, our project explores whether altering memory valence could produce more durable therapeutic outcomes than current behavioural treatments alone. Developing this proposal has highlighted both the scientific potential and ethical complexity of memory-focused interventions, particularly for conditions characterized by unstable self-representation. Through this interdisciplinary integration, our project sheds light on emerging avenues for BPD treatment while emphasizing the need for responsible clinical and philosophical evaluation moving forward.

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